

Nikhil Rajan

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EDUCATION

London School of Economics and Political Science <i>MSc in Financial Statistics</i>	London, UK <i>August 2026 - June 2027</i>
• Coursework: Stochastic Calculus, Mathematics of Market Microstructure, Fixed Income, Computational Methods in Finance, Derivatives, Machine Learning in Finance, Financial Statistics, Advanced Statistics.	
Madras School of Economics <i>Bachelor of Arts (Honours) in Economics</i>	Chennai, India <i>August 2023 - June 2026</i>
• Cumulative Grade Point Average: 9.61/10.0 ♦ Gold Medalist (Rank 1/60) • Dean's List (Distinction): Semesters II - VI • Coursework: Stochastic Processes, Advanced Time Series, Real Analysis, Advanced Differential Equations, Linear Algebra, Econometrics, Statistics, Game Theory, Money and Banking, Data Science, Python and MATLAB.	

INTERNSHIP EXPERIENCE

Madras School of Economics <i>Research Intern (Advisor: Dr. Naveen Srinivasan)</i>	Chennai, India <i>December 2024 - January 2025</i>
• Investigated horizontal innovation and knowledge spillovers in the Romer growth model using analytical and numerical methods. • Contributed to a new micro-founded model linking institutional quality, innovation, and long-run productivity.	
V MACS Yosaney <i>Finance Intern</i>	Chennai, India <i>May 2024 - July 2024</i>
• Built an automated tool for generating P&L statements, balance sheets, and financial schedules, reducing reporting time and errors. • Analyzed MSME financial statements and evaluated bank/NBFC credit products to understand documentation and risk requirements.	

QUANTITATIVE RESEARCH PROJECTS

Liquidity Regimes as a Markov Chain and Optimal Execution as a Markov Decision Process (Ongoing) Modelling liquidity as a four-state Markov process and exploring continuous-time optimal execution under persistent liquidity regimes.
Credit Rating Transitions and Markov Chains Reconstructed Jarrow - Lando - Turnbull (1997) from first principles; derived hedging strategies and risk-neutral transition matrices.
Stochastic Welfare Analysis of Sin Taxation Under India's New GST Regime Modelled cigarette taxation through a Markov-based behavioural transition model using NSSO microdata.
Hidden Markov Model Framework for Credit Stress Detection and Recession Early Warning (Ongoing) Developing a hidden Markov and mean-reverting stochastic process to detect credit stress regimes, quantify their lead-lag relationship with recessions, and drive a backtested tactical allocation strategy.
Causal Analysis of Jump Contagion in Equity Markets: Evidence from Nifty 50 and S&P 500 Returns Quantified Nifty 50 and S&P 500 contagion using multi-model analysis, finding 0.284 overnight transmission and $3.42\times$ crisis volatility.

TEACHING ASSISTANTSHIPS

Foundations of Modern Finance I & II (Course Link) Conducted quizzes and weekly assignments; explained CAPM, Stochastic Discount Factors, and derivatives.	Sem IV
Stochastic Processes Created detailed notes on martingales, Markov Decision Processes, and Brownian motion; led problem-solving sessions.	Sem V
Time Series Analysis Worked on weekly assignments, with the syllabus covering the first five chapters of Hamilton's Time Series Analysis.	Sem VI
Game Theory Engaged with weekly assignments while working through core game theoretic concepts.	Sem VI
Multivariable Calculus Delivered revision lectures on multivariate calculus; held tutorials guiding juniors through assignments and proofs.	Sem II, IV
Intermediate Microeconomics Designed assignments and grading rubrics on monopoly, exchange, and externalities.	Sem V

EXTRACURRICULAR ACHIEVEMENTS

App Development - Eminence Class Recognition • Developed an education-tech application under India's first project-based learning program by IIT Madras, Rhapsody Plus, Intellect, and UC San Diego.
Chess (selected): • Placed 13th on merit list in 1st SRM International Open FIDE Rapid Chess Tournament (2022) • Captained the school team in CBSE Cluster Championships; scored 5.5/6 on Board 1 (2022) • Placed 2nd in Under-15 category in Talent Chess Academy's 2nd FIDE-rated tournament (2019)

SKILLS

• Quantitative & Financial Modeling: Stochastic Processes, Econometrics, Time Series Analysis, Asset Pricing, Game Theory, Market Microstructure, Credit Risk Modeling. • Technical: Python, MATLAB, Julia, Stata, Excel, Latex. • Analytical Foundations: Real Analysis, Statistics, Differential Equations, Linear Algebra.
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